

ARTIFICIAL INTELLIGENCE

Assignment – 1

AI-Based Smart Classroom Attendance System

Submitted By

Thomson Leo Thomas

Group 2

S7 CSEA

Roll No: 59

1 Introduction

The AI-Based Smart Classroom Attendance System is an intelligent software agent that automates student attendance using facial recognition technology. The system captures live video through a webcam, detects and recognizes registered students, and automatically records their attendance in a database. Unlike traditional attendance methods, it reduces manual effort, saves classroom time, and minimizes human errors.

The intelligent agent compares the detected face with previously registered facial encodings and marks attendance only when a valid match is found. It also prevents duplicate attendance records and performs liveness verification to reduce proxy attendance attempts using photographs or mobile screens. Attendance records are securely stored in an SQLite database and can be viewed or exported whenever required.

The system provides an efficient, secure, and intelligent solution for classroom attendance management while improving accuracy, transparency, and overall administrative efficiency.

2 PEAS Representation

Agent Type	Performance Measure	Environment	Actuators	Sensors
AI-Based Smart Classroom Attendance System	High face recognition accuracy, Fast attendance marking, Prevent proxy attendance, No duplicate attendance, Accurate attendance reports	Classroom, Students, Faculty, Webcam, Database	Display recognized student, Mark attendance, Store attendance, Export reports, Generate analytics	Webcam, Student database, Facial encodings, Date & Time, User input

3 Environment Classification

Property	Type	Justification
Observability	Partially Observable	The webcam can only recognize students who are visible within its field of view.
Determinism	Stochastic	Lighting conditions, facial expressions, head movements, and camera noise affect recognition accuracy.
Episode Type	Sequential	Current attendance decisions depend on previous attendance records stored in the database.
Dynamics	Dynamic	Students continuously move, change posture, or leave the camera frame during attendance.
State Space	Discrete	Each student is classified as either Present or Absent.
Agents	Single Agent	Only one AI attendance agent performs recognition and attendance marking.

4 Rationality Measure

A rational agent always selects the action that maximizes its performance based on the information available. The Smart Classroom Attendance System behaves rationally by accurately detecting faces, recognizing registered students, verifying liveness, and recording attendance only when all conditions are satisfied.

The system first detects a student's face using OpenCV and extracts facial features for comparison with stored facial encodings. If the recognition confidence is greater than the predefined threshold, the student is identified successfully. Before recording attendance, the system checks whether attendance has already been marked for the same student to avoid duplicate entries.

To prevent proxy attendance, the system performs liveness detection by verifying eye presence and rejecting photographs or mobile screen images. Unknown faces or faces with low confidence scores are ignored. If the webcam is unavailable or the database cannot be accessed, the system handles these situations gracefully without crashing.

By maximizing recognition accuracy, reducing false positives, preventing duplicate records, and blocking proxy attendance, the AI agent consistently behaves as a rational intelligent system.

5 Innovation

1. AI-Based Anti-Spoofing

The system performs passive liveness detection by checking for eye presence before marking attendance. This prevents attendance being marked using printed photographs or mobile phone images.

2. Real-Time Attendance Dashboard

Attendance data is automatically synchronized to a dashboard where teachers can monitor attendance statistics, student participation, and absentee reports in real time.

3. Automatic Parent Notification

At the end of each attendance session, absent students are identified automatically, and notification messages can be sent to parents through Email, SMS, or WhatsApp to improve communication between the institution and parents.

6 System Architecture

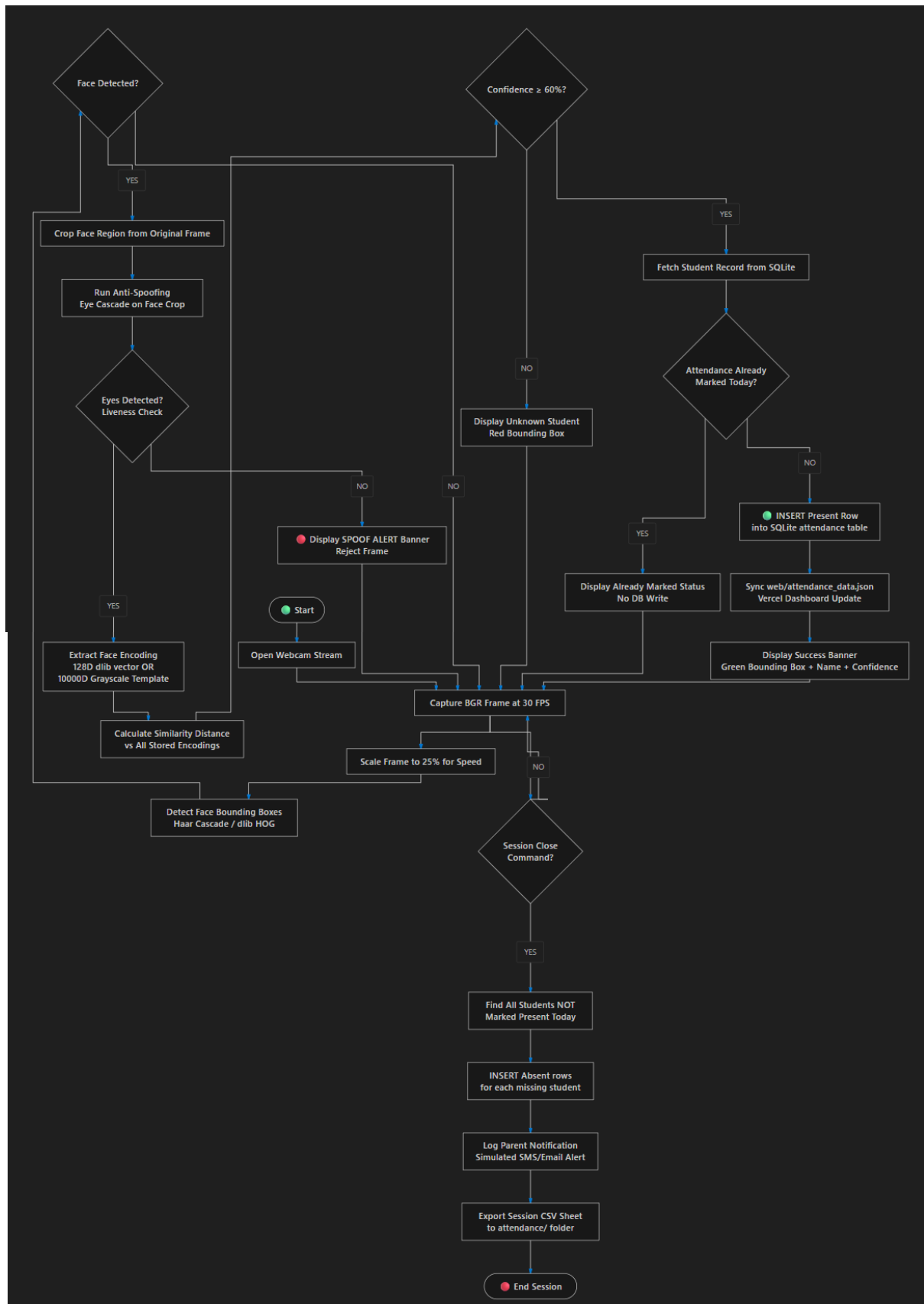


Figure 1: System Architecture of the AI-Based Smart Classroom Attendance System

7 Conclusion

The AI-Based Smart Classroom Attendance System demonstrates an effective application of Artificial Intelligence and Computer Vision in educational institutions. The system automates attendance recording, reduces manual effort, improves accuracy, and minimizes proxy attendance through facial recognition and liveness verification. With features such as duplicate prevention, attendance analytics, and parent notifications, the proposed system provides a reliable, secure, and intelligent attendance management solution for modern classrooms.

8 Simulation

The simulation of the **AI-Based Smart Classroom Attendance System** demonstrates the complete workflow of the intelligent agent through a command-line interface (CLI) integrated with computer vision. Initially, the administrator launches the application and selects the required operation from the main menu, such as student registration, model training, attendance marking, viewing attendance logs, or exporting records.

For student registration, the administrator enters the student's roll number, name, department, and semester. The webcam is then activated to capture multiple facial images of the student. These images are stored in the dataset and later used to train the face recognition model.

After registration, the administrator trains the face recognition model. The system processes all captured face images, generates facial templates, and stores the trained model for future recognition. This enables the intelligent agent to accurately identify registered students during attendance sessions.

During the live attendance session, the administrator enters the subject name and activates the webcam. The AI agent continuously detects faces from the live video stream, verifies liveness, and compares the detected face with the stored facial templates. When a valid match is found with sufficient confidence, the system automatically marks the student's attendance and stores the student's name, roll number, date, time, subject, and attendance status in the SQLite database. Duplicate attendance entries are prevented by checking whether attendance has already been marked for the same session.

After the attendance session is completed, the system automatically identifies absent students, records their attendance status as **Absent**, generates parent notification messages, and exports the complete attendance report as a CSV file. The administrator can also view all attendance records through the CLI menu and export them for future reference.

The simulation demonstrates that the proposed AI agent performs accurate facial recognition, prevents duplicate attendance, maintains digital attendance records, and automates classroom attendance management with minimal human intervention.

9 Screenshots of Full Code

The following figures present the important code snippets of the AI-Based Smart Classroom Attendance System. The code implements student registration, face recognition, live attendance marking, attendance management, and report generation using Python and OpenCV.

```
8 import os
9 import sys
10 import cv2
11 import time
12 import argparse
13 import tkinter as tk
14 from tkinter import ttk, messagebox
15 from PIL import Image, ImageTk
16 from datetime import datetime
17
18 # local imports
19 import database
20 import utils
21 import register
22 import train_model
23 from recognizer import SmartRecognizer
24 import attendance
25
26 # Ensure DB and directories are ready
27 database.init_db()
28 utils.create_required_directories()
29
30
```

Figure 2: Main Program Initialization and Module Import

Importing required Python libraries, initializing the database, and loading project modules required for the Smart Classroom Attendance System.

```
31 # -----
32 # PART 1: CLI INTERFACE
33 # -----
34
35 def run_cli_attendance():
36     """
37     Runs a live attendance session from the terminal.
38     Opens OpenCV window and marks attendance in real-time.
39     """
40     print("\n" + "-" * 40)
41     print("    LIVE ATTENDANCE SESSION (CLI)    ")
42     print("-" * 40)
43
44     subject = input("Enter Subject Name (e.g., AI_Lab): ").strip()
45     if not subject:
46         print("[ERROR] Subject name is required.")
47         return
48
49     print("\nLoading face recognition models...")
50     recognizer = SmartRecognizer()
51     if not recognizer.known_names:
52         print("[ERROR] No face encodings found. Please register students and run training first.")
53         return
54
55     print("\nStarting webcam...")
56     cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)
57     if not cap.isOpened():
58         cap = cv2.VideoCapture(0)
59         if not cap.isOpened():
60             print("[ERROR] Could not open webcam.")
61             return
62
63     print("\nWebcam Active. Standing by to scan faces...")
64     print("Press 'Q' to close the attendance session.")
65
66     last_checked = {} # Throttle marking frequency
```

Figure 3: Live Attendance Session Function

Implementation of the live attendance session, including webcam initialization, subject selection, and preparation of the face recognition system.

```

68 while True:
69     ret, frame = cap.read()
70     if not ret:
71         print("[ERROR] Failed to grab frame.")
72         break
73
74     # Get recognition results
75     faces = recognizer.recognize(frame)
76
77     for face in faces:
78         top, right, bottom, left = face["box"]
79         name = face["name"]
80         roll = face["roll"]
81         conf = face["confidence"]
82         is_live = face["liveness"]
83
84         # Set color: Red for spoof/unknown, Green for recognized live face
85         if name == "Unknown":
86             color = (0, 0, 255) # Red
87             label = f"Unknown"
88         elif not is_live:
89             color = (0, 165, 255) # Orange (Spoof Alert)
90             label = f"SPOOF ALERT! {name}"
91         else:
92             color = (0, 255, 0) # Green
93             label = f"{name} {roll} {conf}%"
94
95         # Throttle attendance mark to once every 5 seconds per student in frame
96         now_ts = time.time()
97         if roll not in last_checked or (now_ts - last_checked[roll]) > 5:
98             last_checked[roll] = now_ts
99             marked, msg = attendance.mark_student_present(roll, subject)
100             print(f"[ATTENDANCE] {msg}")
101

```

Figure 4: Face Recognition and Attendance Marking
Recognition of student faces, liveness verification, confidence calculation, and automatic attendance marking using facial recognition.

```

102 # Draw box and label
103 cv2.rectangle(frame, (left, top), (right, bottom), color, 2)
104 cv2.putText(frame, label, (left, top - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.6, color, 2)
105
106 # Draw liveness flag
107 liveness_text = "Liveness: Verified" if is_live else "Liveness: FAILED (Static Image)"
108 cv2.putText(frame, liveness_text, (left, bottom + 20), cv2.FONT_HERSHEY_SIMPLEX, 0.5, color, 1)
109
110 cv2.imshow(f"Marking Attendance: {subject}", frame)
111
112 # Press Q to quit
113 if cv2.waitKey(1) & 0xFF == ord("q"):
114     break
115
116 cap.release()
117 cv2.destroyAllWindows()
118
119 # Automatically trigger absent markings and parent notifications
120 print(f"[*]Closing attendance session for '{subject}'...")
121 absent_count, absents = attendance.close_attendance_session(subject)
122 print(f"[*]SESSION CLOSED] Marked {absent_count} students as Absent today.")
123 if absent_count > 0:
124     print("Absents List:")
125     for stu in absents:
126         print(f"  {stu['name']} {stu['roll_number']}")
127     print("Exporting report...")
128     csv_path = utils.export_attendance_to_csv(datetime.now().strftime("%Y-%m-%d"))
129     print(f"[*]SUCCESS] Today's attendance sheet exported to: {csv_path}")
130

```

Figure 5: Attendance Session Completion
Closing the attendance session, marking absent students, generating notifications, and exporting the attendance report.


```

132  def view_reports_cli():
133      """
134      Displays attendance records in the terminal.
135      """
136      print("\n" + "-" * 40)
137      print("      ATTENDANCE LOGS (CLI)      ")
138      print("-" * 40)
139      records = database.get_all_attendance()
140      if not records:
141          print("No attendance records found.")
142          return
143
144      print(f"{'ID':<5} | {'Roll No':<10} | {'Name':<20} | {'Date':<12} | {'Time':<10} | {'Subject':<12} | {'Status':<8}")
145      print("-" * 85)
146      for r in records:
147          print(f"{r['attendance_id']:<5} | {r['roll_number']:<10} | {r['student_name']:<20} | {r['date']:<12} | {r['time']:<10} | {r['subject']:<12} | {r['status']:<8}")
148
149
150  def run_cli_menu():
151      """
152      Main loop for interactive terminal mode.
153      """
154      while True:
155          print("\n" + "-" * 50)
156          print("  AI-BASED SMART CLASSROOM ATTENDANCE SYSTEM  ")
157          print("-" * 50)
158          print("1. Register New Student")
159          print("2. Train AI Face Recognition Model")
160          print("3. Start Live Attendance Session")
161          print("4. View Attendance Logs")
162          print("5. Export All Records to CSV")
163          print("6. Exit")
164          print("-" * 50)
165
166          choice = input("Enter choice (1-6): ").strip()
167

```

Figure 6: Attendance Log Management

Displaying attendance records stored in the database, including student details, date, time, subject, and attendance status.

```

165
166     choice = input("Enter choice (1-6): ").strip()
167
168     if choice == "1":
169         register.register_student_cli()
170     elif choice == "2":
171         train_model.run_training()
172     elif choice == "3":
173         run_cli_attendance()
174     elif choice == "4":
175         view_reports_cli()
176     elif choice == "5":
177         csv_path = utils.export_attendance_to_csv()
178         print(f"[SUCCESS] CSV Exported to: {csv_path}")
179     elif choice == "6":
180         print("\nExiting. Thank you for using Smart Classroom Attendance!")
181         sys.exit(0)
182     else:
183         print("[ERROR] Invalid selection. Choose between 1 and 6.")
184
185

```

Figure 7: Main CLI Menu

Implementation of the Command Line Interface (CLI) menu providing options for student registration, model training, attendance marking, viewing attendance logs, and exporting reports.

10 Output

The following figures show the execution and output of the AI-Based Smart Classroom Attendance System at different stages, including student registration, face recognition, attendance marking, attendance log generation, and report export.

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS D:\Smart Classroom Attendance System> git push -u origin main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 12 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (21/21), 41.53 KiB | 867.00 KiB/s, done.
Total 21 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/thomson-leo-thomas/Smart-Attendance.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\Smart Classroom Attendance System> python main.py --cli

=====
AI-BASED SMART CLASSROOM ATTENDANCE SYSTEM
=====
1. Register New Student
2. Train AI Face Recognition Model
3. Start Live Attendance Session
4. View Attendance Logs
5. Export All Records to CSV
6. Exit
=====
Enter choice (1-6):

```

Figure 8: Main Menu

The application starts with a Command Line Interface (CLI) that provides options to register new students, train the face recognition model, start a live attendance session, view attendance logs, export attendance reports, and exit the application.

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS D:\Smart Classroom Attendance System> git push -u origin main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 12 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (21/21), 41.53 KiB | 867.00 KiB/s, done.
Total 21 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/thomson-leo-thomas/Smart-Attendance.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\Smart Classroom Attendance System> python main.py --cli

=====
AI-BASED SMART CLASSROOM ATTENDANCE SYSTEM
=====
1. Register New Student
2. Train AI Face Recognition Model
3. Start Live Attendance Session
4. View Attendance Logs
5. Export All Records to CSV
6. Exit
=====
Enter choice (1-6): 1

=====
STUDENT REGISTRATION MENU (CLI)
=====
Enter Roll Number: 59
Enter Full Name: Thomson Leo Thomas
Enter Department: cse
Enter Semester: s7

Preparing to capture face photos...
Webcam will open. Look directly into the lens.
Press SPACEBAR to capture an image. We need 10 samples.

[INFO] Starting camera for: Thomson Leo Thomas. Look at the camera.

```

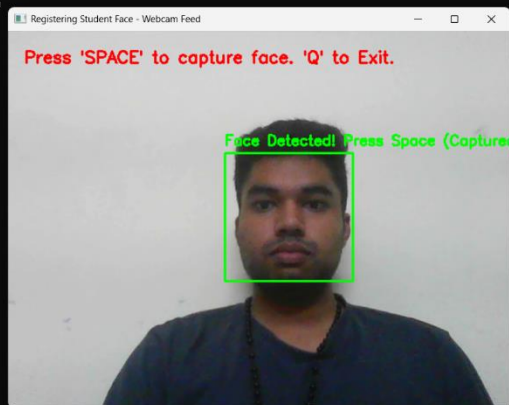


Figure 9: Student Registration

The administrator enters the student's roll number, name, department, and semester. The webcam captures multiple face samples, which are stored in the dataset for training the recognition model.

```
3. Start Live Attendance Session
4. View Attendance Logs
5. Export All Records to CSV
6. Exit
=====
Enter choice (1-6): 1

=====
STUDENT REGISTRATION MENU (CLI)
=====
Enter Roll Number: 59
Enter Full Name: Thomson Leo Thomas
Enter Department: cse
Enter Semester: s7

Preparing to capture face photos...
Webcam will open. Look directly into the lens.
Press SPACEBAR to capture an image. We need 10 samples.

[INFO] Starting camera for: Thomson Leo Thomas. Look at the camera.
[INFO] Captured 1/10 successfully!
[INFO] Captured 2/10 successfully!
[INFO] Captured 3/10 successfully!
[INFO] Captured 4/10 successfully!
[INFO] Captured 5/10 successfully!
[INFO] Captured 6/10 successfully!
[INFO] Captured 7/10 successfully!
[INFO] Captured 8/10 successfully!
[INFO] Captured 9/10 successfully!
[INFO] Captured 10/10 successfully!

[SUCCESS] Student Thomson Leo Thomas registered in DB.
Successfully captured 10 face samples.
Note: Remember to run Model Training to compile new faces!

=====
AI-BASED SMART CLASSROOM ATTENDANCE SYSTEM
=====
1. Register New Student
2. Train AI Face Recognition Model
3. Start Live Attendance Session
4. View Attendance Logs
5. Export All Records to CSV
6. Exit
=====
Enter choice (1-6): |
```

Figure 10: Student Registration Completed

After capturing ten face samples, the system successfully stores the student's information in the database and confirms successful registration. The administrator is then instructed to train the recognition model.

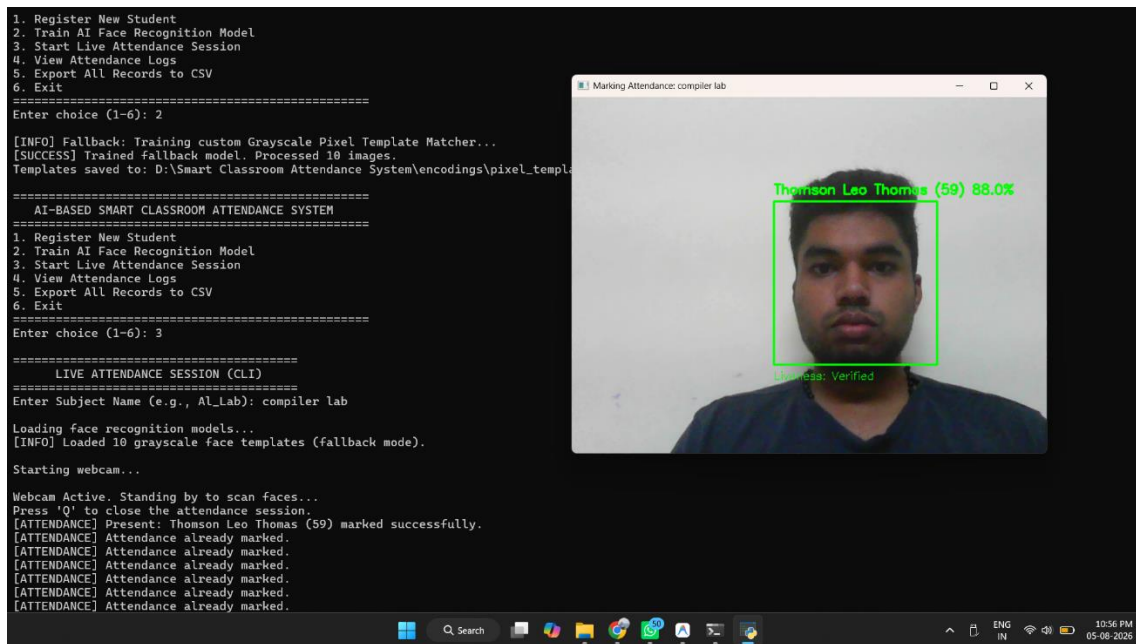


Figure 11: Live Attendance Session

The trained model detects and recognizes the student's face through the webcam. Once the confidence threshold is satisfied and liveness is verified, attendance is automatically marked and stored in the database.

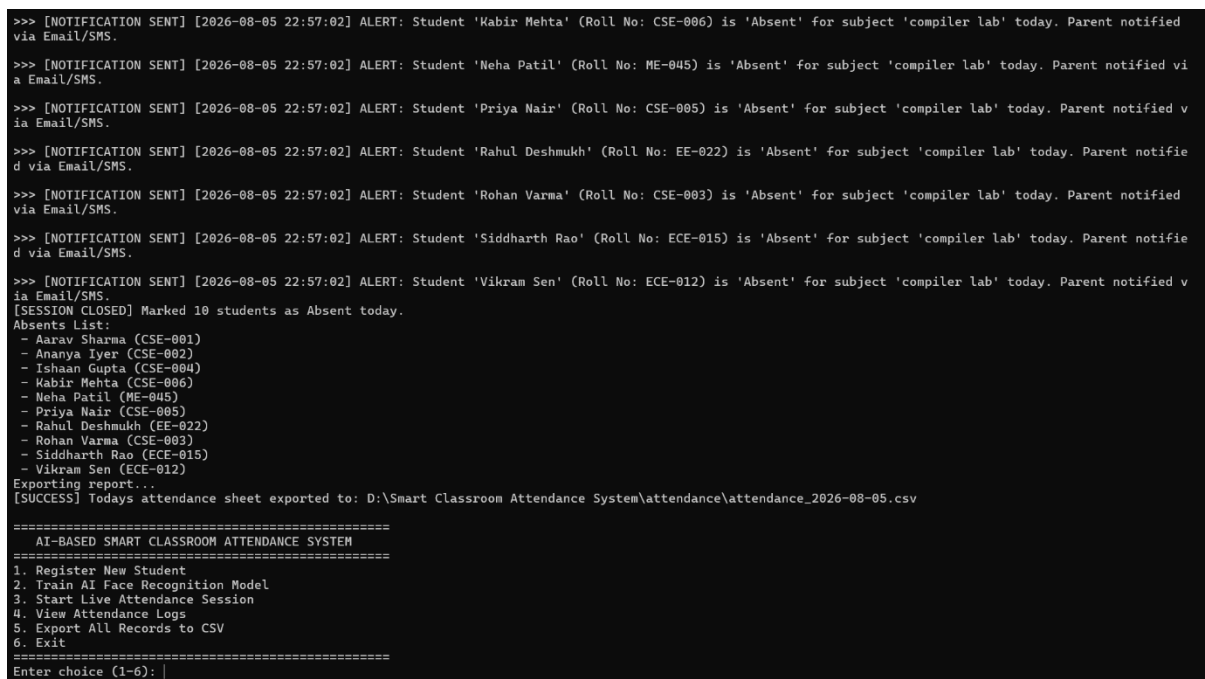


Figure 12: Attendance Completion and Notifications

At the end of the attendance session, absent students are automatically identified. The system generates notification messages for parents and exports the attendance report as a CSV file.

```
=====
AI-BASED SMART CLASSROOM ATTENDANCE SYSTEM
=====
1. Register New Student
2. Train AI Face Recognition Model
3. Start Live Attendance Session
4. View Attendance Logs
5. Export All Records to CSV
6. Exit
=====
Enter choice (1-6): 4
=====
ATTENDANCE LOGS (CLI)
=====
ID | Roll No | Name | Date | Time | Subject | Status
-----
92 | CSE-001 | Aarav Sharma | 2026-08-05 | 22:57:02 | compiler lab | Absent
93 | CSE-002 | Ananya Iyer | 2026-08-05 | 22:57:02 | compiler lab | Absent
94 | CSE-004 | Ishaan Gupta | 2026-08-05 | 22:57:02 | compiler lab | Absent
95 | CSE-006 | Kabir Mehta | 2026-08-05 | 22:57:02 | compiler lab | Absent
96 | ME-045 | Neha Patil | 2026-08-05 | 22:57:02 | compiler lab | Absent
97 | CSE-005 | Priya Nair | 2026-08-05 | 22:57:02 | compiler lab | Absent
98 | EE-022 | Rahul Deshmukh | 2026-08-05 | 22:57:02 | compiler lab | Absent
99 | CSE-003 | Rohan Varma | 2026-08-05 | 22:57:02 | compiler lab | Absent
100 | ECE-015 | Siddharth Rao | 2026-08-05 | 22:57:02 | compiler lab | Absent
101 | ECE-012 | Vikram Sen | 2026-08-05 | 22:57:02 | compiler lab | Absent
91 | S9 | Thomson Leo Thomas | 2026-08-05 | 22:56:06 | compiler lab | Present
1 | CSE-001 | Aarav Sharma | 2026-08-05 | 10:15:00 | AI_Lab | Absent
8 | ECE-015 | Siddharth Rao | 2026-08-05 | 10:15:00 | AI_Lab | Absent
11 | CSE-001 | Aarav Sharma | 2026-08-05 | 10:15:00 | Computer_Vision | Absent
18 | ECE-015 | Siddharth Rao | 2026-08-05 | 10:15:00 | Computer_Vision | Absent
21 | CSE-001 | Aarav Sharma | 2026-08-05 | 10:15:00 | Deep_Learning | Absent
28 | ECE-015 | Siddharth Rao | 2026-08-05 | 10:15:00 | Deep_Learning | Absent
2 | CSE-002 | Ananya Iyer | 2026-08-05 | 09:05:00 | AI_Lab | Present
3 | CSE-003 | Rohan Varma | 2026-08-05 | 09:05:00 | AI_Lab | Present
4 | CSE-004 | Ishaan Gupta | 2026-08-05 | 09:05:00 | AI_Lab | Present
5 | CSE-005 | Priya Nair | 2026-08-05 | 09:05:00 | AI_Lab | Present
6 | CSE-006 | Kabir Mehta | 2026-08-05 | 09:05:00 | AI_Lab | Present
7 | ECE-012 | Vikram Sen | 2026-08-05 | 09:05:00 | AI_Lab | Present
9 | ME-045 | Neha Patil | 2026-08-05 | 09:05:00 | AI_Lab | Present
10 | EE-022 | Rahul Deshmukh | 2026-08-05 | 09:05:00 | AI_Lab | Present
12 | CSE-002 | Ananya Iyer | 2026-08-05 | 09:05:00 | Computer_Vision | Present
13 | CSE-003 | Rohan Varma | 2026-08-05 | 09:05:00 | Computer_Vision | Present
14 | CSE-004 | Ishaan Gupta | 2026-08-05 | 09:05:00 | Computer_Vision | Present
```

Figure 13: Attendance Logs

The administrator can view all attendance records through the CLI, including the attendance ID, student details, date, time, subject, and attendance status.

```
71 | CSE-001 | Aarav Sharma | 2026-08-03 | 09:05:00 | Computer_Vision | Present
72 | CSE-002 | Ananya Iyer | 2026-08-03 | 09:05:00 | Computer_Vision | Present
73 | CSE-003 | Rohan Varma | 2026-08-03 | 09:05:00 | Computer_Vision | Present
74 | CSE-004 | Ishaan Gupta | 2026-08-03 | 09:05:00 | Computer_Vision | Present
75 | CSE-005 | Priya Nair | 2026-08-03 | 09:05:00 | Computer_Vision | Present
77 | ECE-012 | Vikram Sen | 2026-08-03 | 09:05:00 | Computer_Vision | Present
78 | ECE-015 | Siddharth Rao | 2026-08-03 | 09:05:00 | Computer_Vision | Present
79 | ME-045 | Neha Patil | 2026-08-03 | 09:05:00 | Computer_Vision | Present
80 | EE-022 | Rahul Deshmukh | 2026-08-03 | 09:05:00 | Computer_Vision | Present
81 | CSE-001 | Aarav Sharma | 2026-08-03 | 09:05:00 | Deep_Learning | Present
82 | CSE-002 | Ananya Iyer | 2026-08-03 | 09:05:00 | Deep_Learning | Present
83 | CSE-003 | Rohan Varma | 2026-08-03 | 09:05:00 | Deep_Learning | Present
84 | CSE-004 | Ishaan Gupta | 2026-08-03 | 09:05:00 | Deep_Learning | Present
85 | CSE-005 | Priya Nair | 2026-08-03 | 09:05:00 | Deep_Learning | Present
87 | ECE-012 | Vikram Sen | 2026-08-03 | 09:05:00 | Deep_Learning | Present
88 | ECE-015 | Siddharth Rao | 2026-08-03 | 09:05:00 | Deep_Learning | Present
89 | ME-045 | Neha Patil | 2026-08-03 | 09:05:00 | Deep_Learning | Present
90 | EE-022 | Rahul Deshmukh | 2026-08-03 | 09:05:00 | Deep_Learning | Present

=====
AI-BASED SMART CLASSROOM ATTENDANCE SYSTEM
=====
1. Register New Student
2. Train AI Face Recognition Model
3. Start Live Attendance Session
4. View Attendance Logs
5. Export All Records to CSV
6. Exit
=====
Enter choice (1-6): 5
[SUCCESS] CSV Exported to: D:\Smart Classroom Attendance System\attendance\attendance_all_20260805_225846.csv
=====
AI-BASED SMART CLASSROOM ATTENDANCE SYSTEM
=====
1. Register New Student
2. Train AI Face Recognition Model
3. Start Live Attendance Session
4. View Attendance Logs
5. Export All Records to CSV
6. Exit
=====
Enter choice (1-6): 6
Exiting. Thank you for using Smart Classroom Attendance!
PS D:\Smart Classroom Attendance System> |
```

Figure 14: CSV Export

The attendance records are successfully exported to a CSV file, allowing the administrator to maintain digital attendance reports for future reference.

Attendance Roll Numb	Student Name	Date	Time	Subject	Status
92 CSE-001	Aarav Sharma	05-08-2026	22:57:02	compiler li	Absent
93 CSE-002	Ananya Iyer	05-08-2026	22:57:02	compiler li	Absent
94 CSE-004	Ishaan Gupta	05-08-2026	22:57:02	compiler li	Absent
95 CSE-006	Kabir Mehta	05-08-2026	22:57:02	compiler li	Absent
96 ME-045	Neha Patil	05-08-2026	22:57:02	compiler li	Absent
97 CSE-005	Priya Nair	05-08-2026	22:57:02	compiler li	Absent
98 EE-022	Rahul Deshmukh	05-08-2026	22:57:02	compiler li	Absent
99 CSE-003	Rohan Varma	05-08-2026	22:57:02	compiler li	Absent
100 ECE-015	Siddharth Rao	05-08-2026	22:57:02	compiler li	Absent
101 ECE-012	Vikram Sen	05-08-2026	22:57:02	compiler li	Absent
91	59 Thomson Leo Thomas	05-08-2026	22:56:06	compiler li	Present
1 CSE-001	Aarav Sharma	05-08-2026	10:15:00	AI_Lab	Absent
8 ECE-015	Siddharth Rao	05-08-2026	10:15:00	AI_Lab	Absent
11 CSE-001	Aarav Sharma	05-08-2026	10:15:00	Computer	Absent
18 ECE-015	Siddharth Rao	05-08-2026	10:15:00	Computer	Absent
21 CSE-001	Aarav Sharma	05-08-2026	10:15:00	Deep_Lear	Absent
28 ECE-015	Siddharth Rao	05-08-2026	10:15:00	Deep_Lear	Absent
2 CSE-002	Ananya Iyer	05-08-2026	09:05:00	AI_Lab	Present
3 CSE-003	Rohan Varma	05-08-2026	09:05:00	AI_Lab	Present
4 CSE-004	Ishaan Gupta	05-08-2026	09:05:00	AI_Lab	Present
5 CSE-005	Priya Nair	05-08-2026	09:05:00	AI_Lab	Present
6 CSE-006	Kabir Mehta	05-08-2026	09:05:00	AI_Lab	Present
7 ECE-012	Vikram Sen	05-08-2026	09:05:00	AI_Lab	Present
9 ME-045	Neha Patil	05-08-2026	09:05:00	AI_Lab	Present
10 EE-022	Rahul Deshmukh	05-08-2026	09:05:00	AI_Lab	Present

Figure 15: Attendance Report in Microsoft Excel

The exported CSV file is opened in Microsoft Excel, displaying all attendance records in a structured tabular format for easy analysis, reporting, and record management.